Data Visualization with Excel

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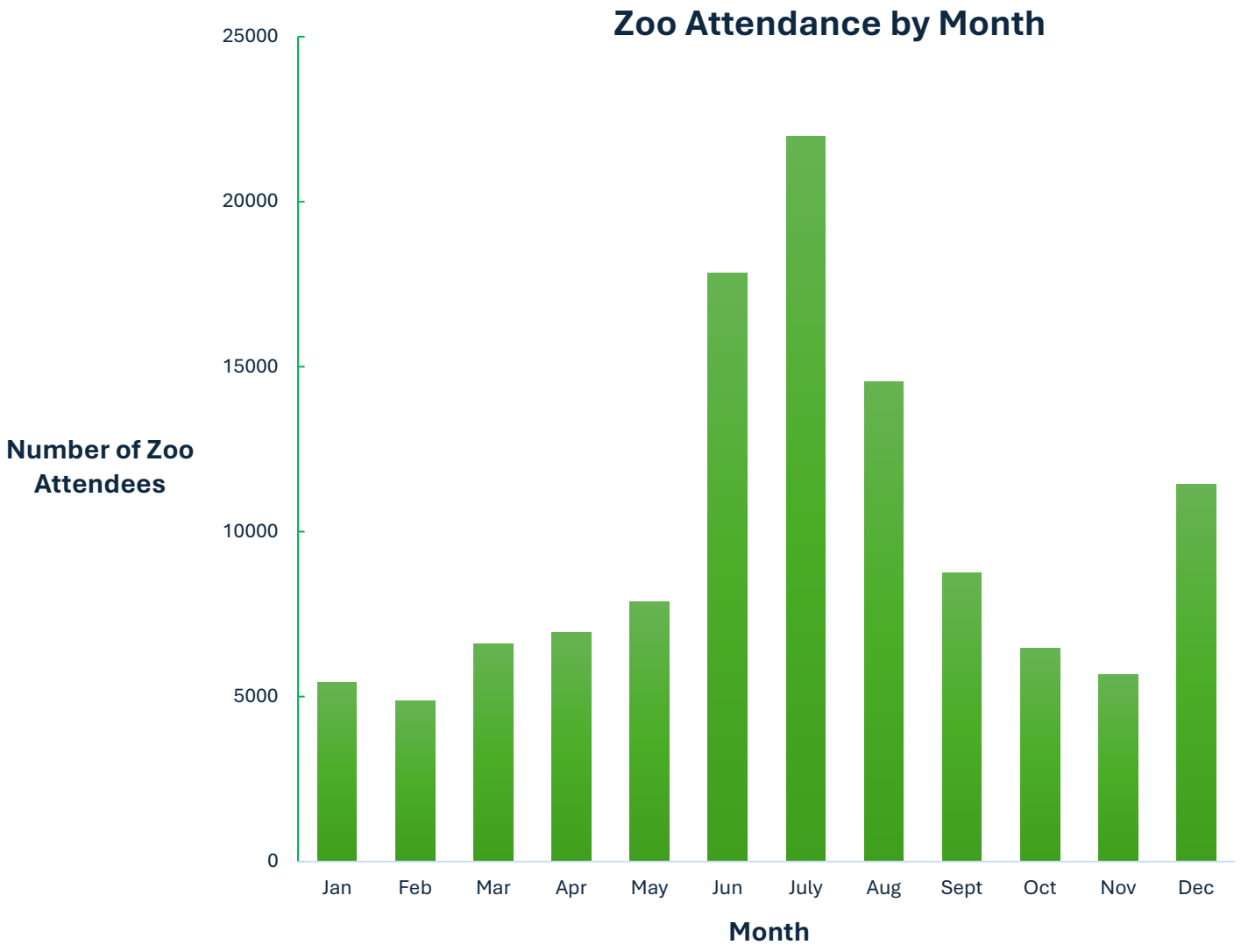
Column Chart (1)

Interpretation: This column chart shows the number of people who went to the zoo each month throughout the year.

Insights

- The zoo has more attendees during the summer months
- Outside of the summer months, the zoo has higher-than-normal attendance during the month of December

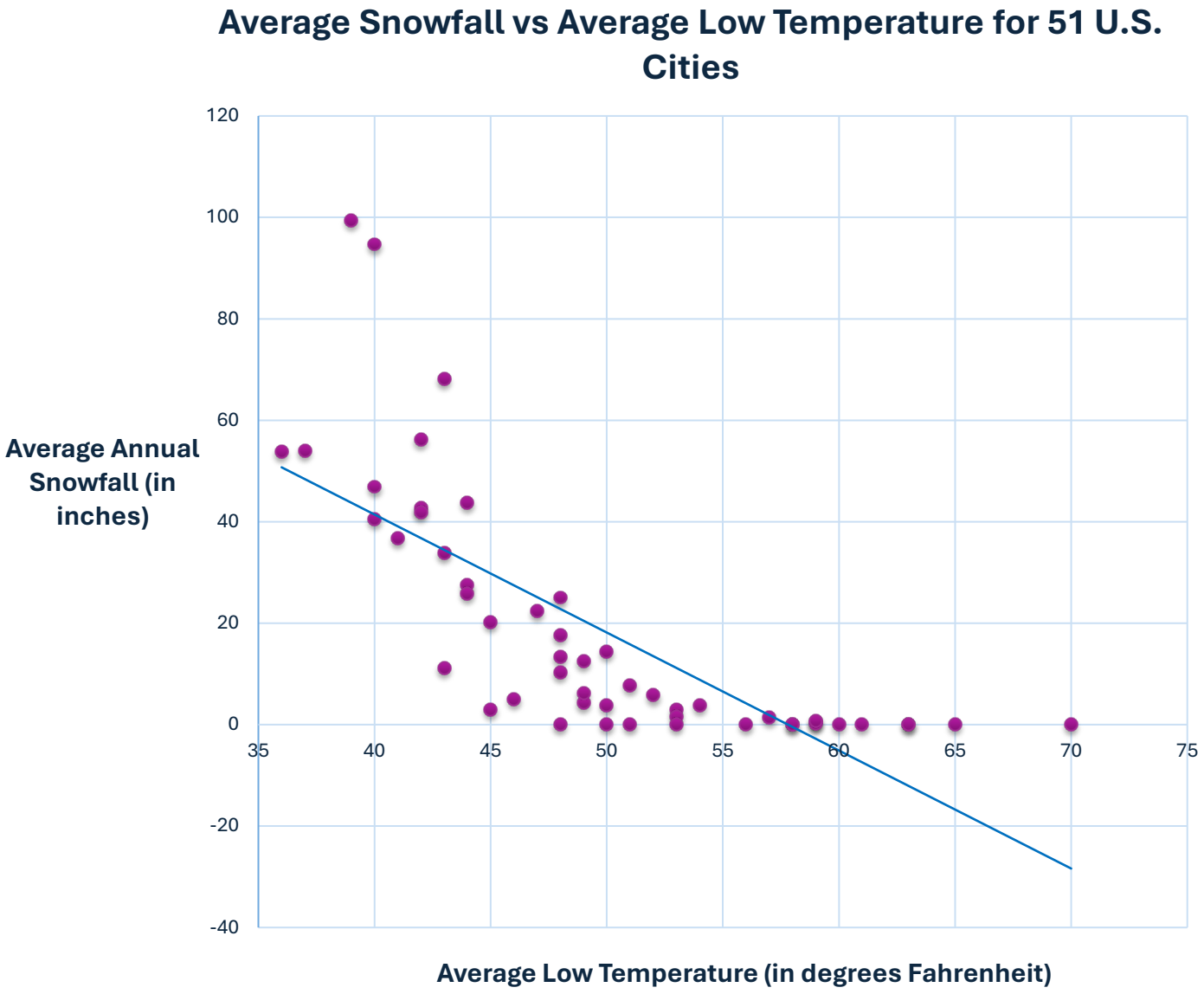


Scatter Chart

Interpretation: This scatter plot shows the relationship between the average low temperature and the average snowfall of 51 cities in the United States.

Insights

- There is a strong negative correlation between average low temperature and average snowfall
    - Cites with lower average temperatures have higher average snowfall and cities with higher average temperatures have lower average snowfall

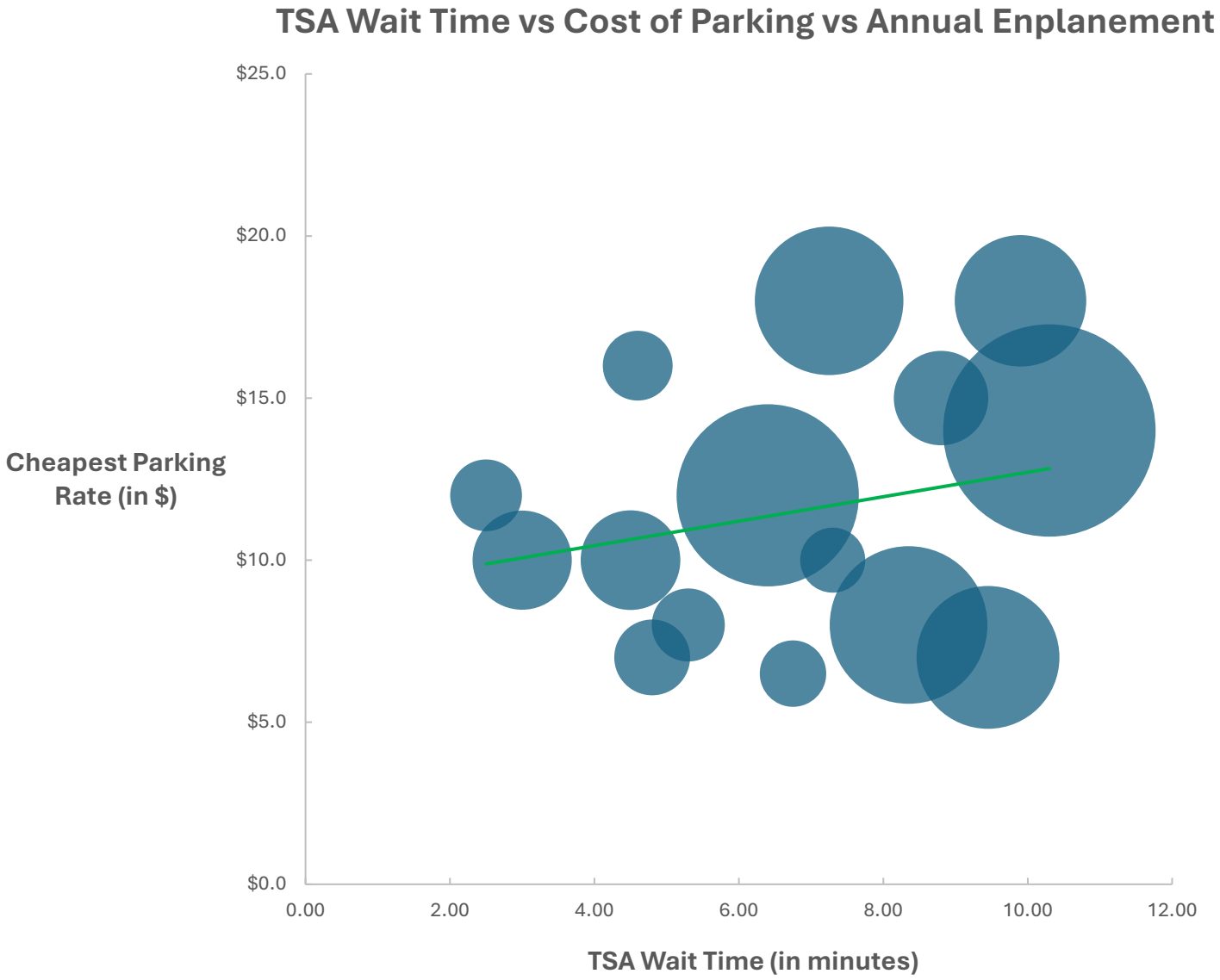


Bubble Chart

Interpretation: This bubble graph shows the relationship between average TSA wait time, the cheapest cost of daily on-site parking rate, and the annual enplanement. The annual enplanement is represented by the size of the bubbles.

Insights

- There seems to be a weak positive correlation between TSA wait time and cheapest parking rate
- TSA wait time is longer when there are more passengers at the airport

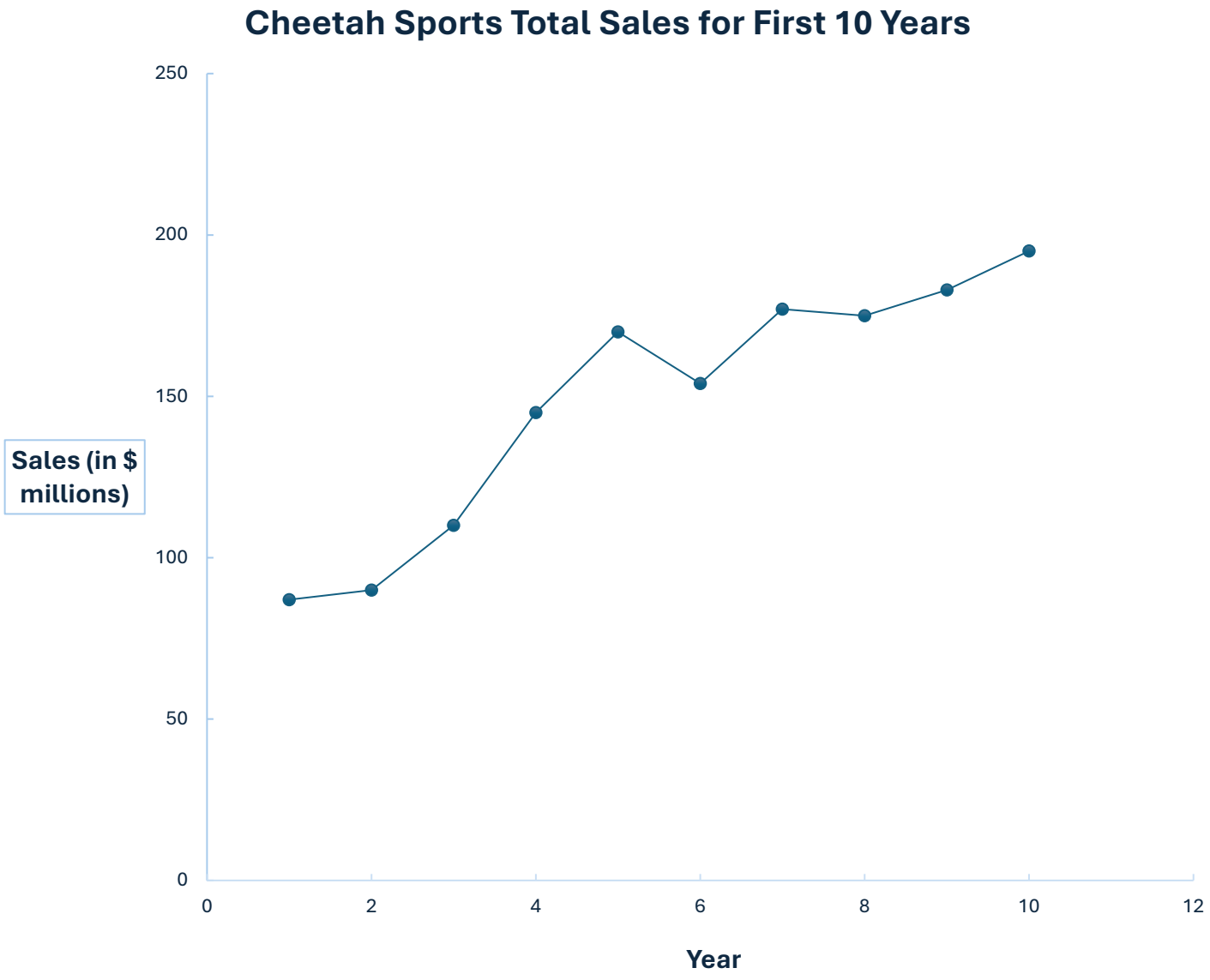


Line Chart

Interpretation: This line chart shows the amount of money Cheetah Sports generated via sales during the first 10 years.

Insights

- Steady increase in sales during the first 5 years
    - Minimal increase in sales between years 1 and 2
- Drop in sales between years 5 and 6
- After year 6, sales start to increase again, but not as much as the first 5 years

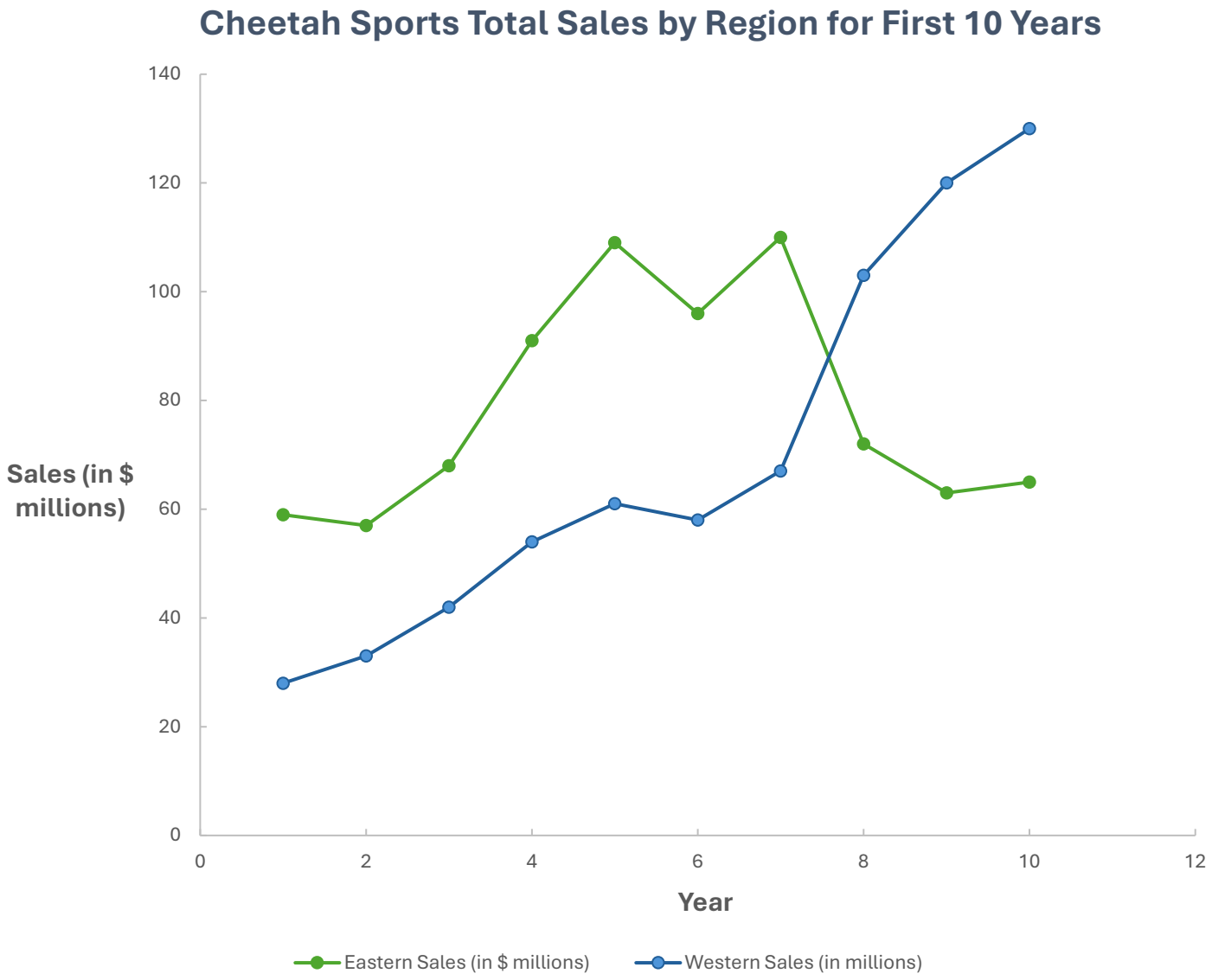


Multiple-Line Chart

Interpretation: This multiple-line chart shows the amount of money Cheetah Sports generated via sales in their eastern and western regions during the first 10 years

Insights

- Before year 8, the eastern region had higher sales than the western region
- From year 7 to year 8, there was a great decline in sales in the eastern region and a great increase in sales in the western region
- After year 8, the western region had higher sales than the eastern region

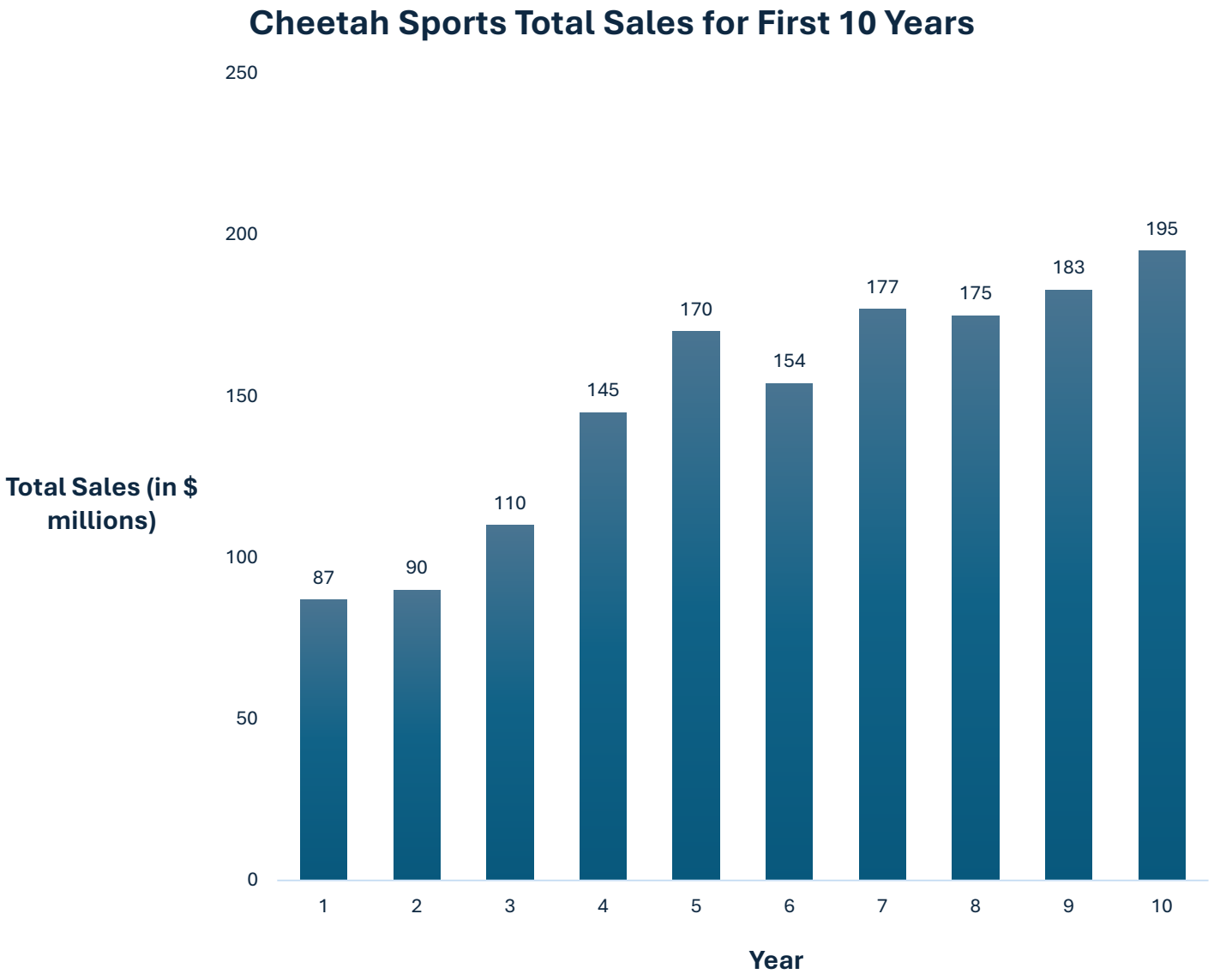


Column Chart (2)

Interpretation: This column chart provides a different way to visualize the amount of money Cheetah Sports generated via sales during the first 10 years.

Insights

- Steady increase in sales during the first 5 years
    - Minimal increase in sales between years 1 and 2
- Drop in sales between years 5 and 6
- After year 6, sales start to increase again, but not as much as the first 5 years

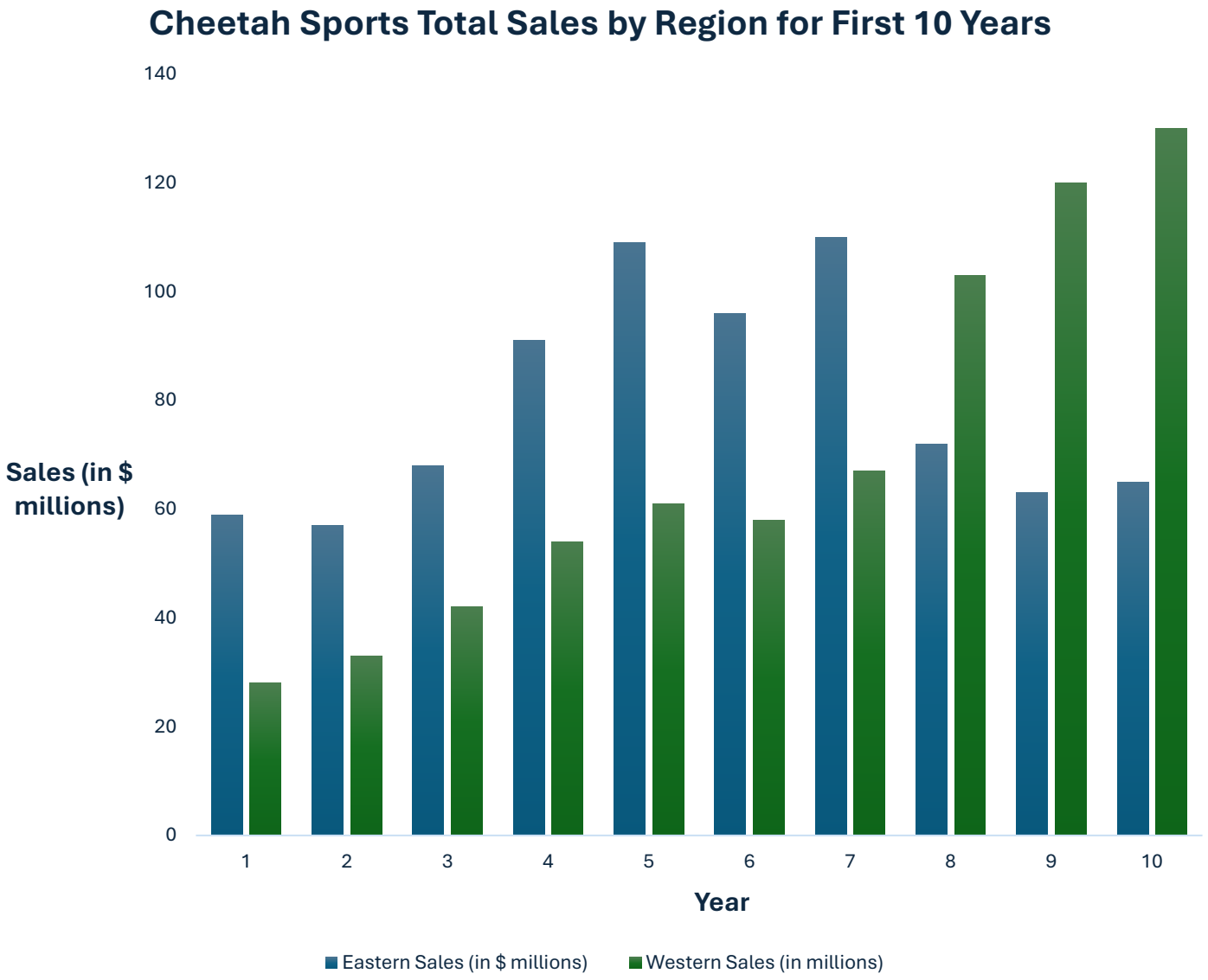


Clustered Column Chart

Interpretation: This clustered column chart shows the amount of money Cheetah Sports generated via sales in their eastern and western regions during the first 10 years.

Insights

- Before year 8, the eastern region had higher sales than the western region
- From year 7 to year 8, there was a great decline in sales in the eastern region and a great increase in sales in the western region
- After year 8, the western region had higher sales than the eastern region

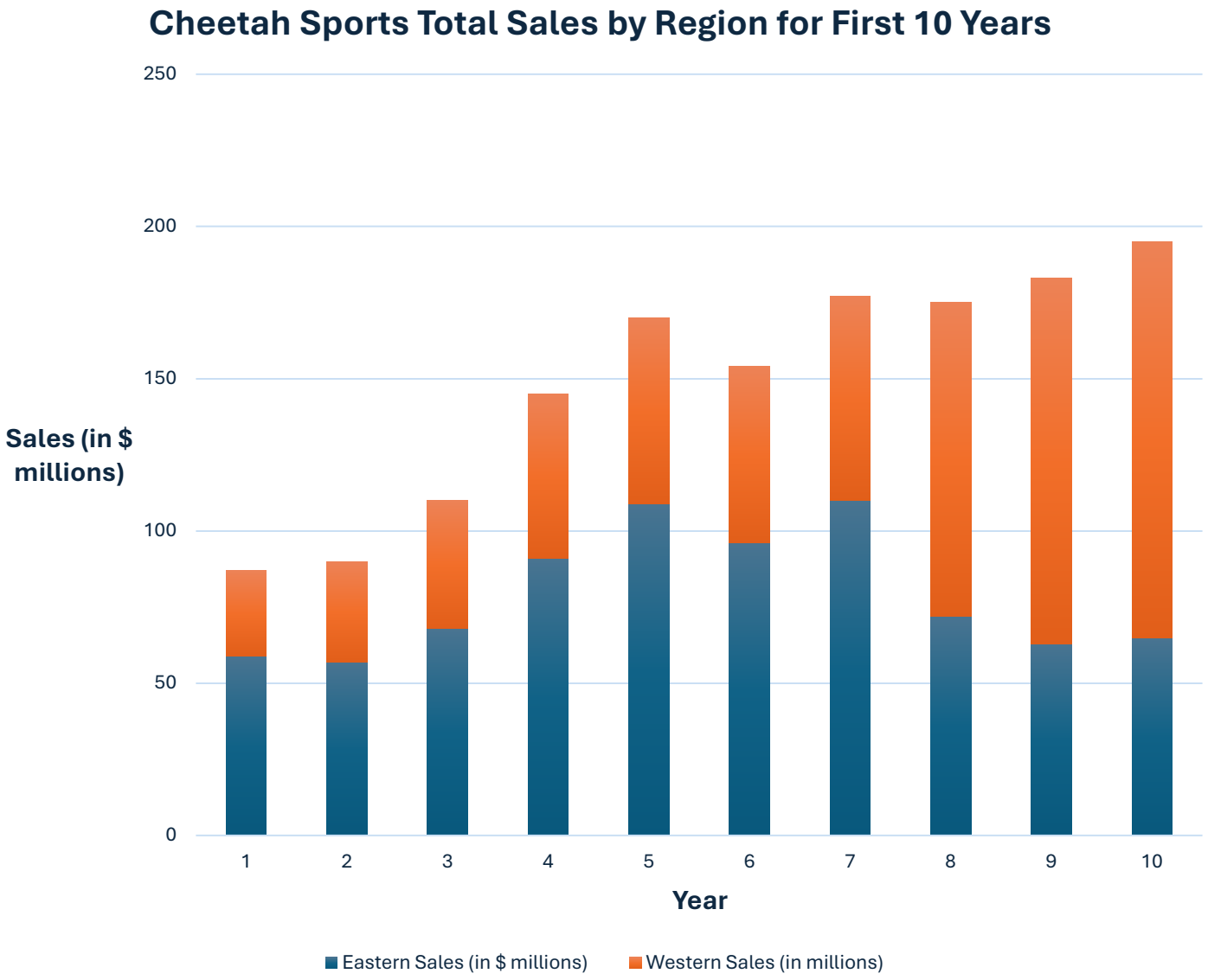


Stacked Column Chart

Interpretation: This stacked column chart shows the amount of money Cheetah Sports generated via sales in their eastern and western regions during the first 10 years.

Insights

- Before year 8, the eastern region had higher sales than the western region
- From year 7 to year 8, there was a great decline in sales in the eastern region and a great increase in sales in the western region
- After year 8, the western region had higher sales than the eastern region

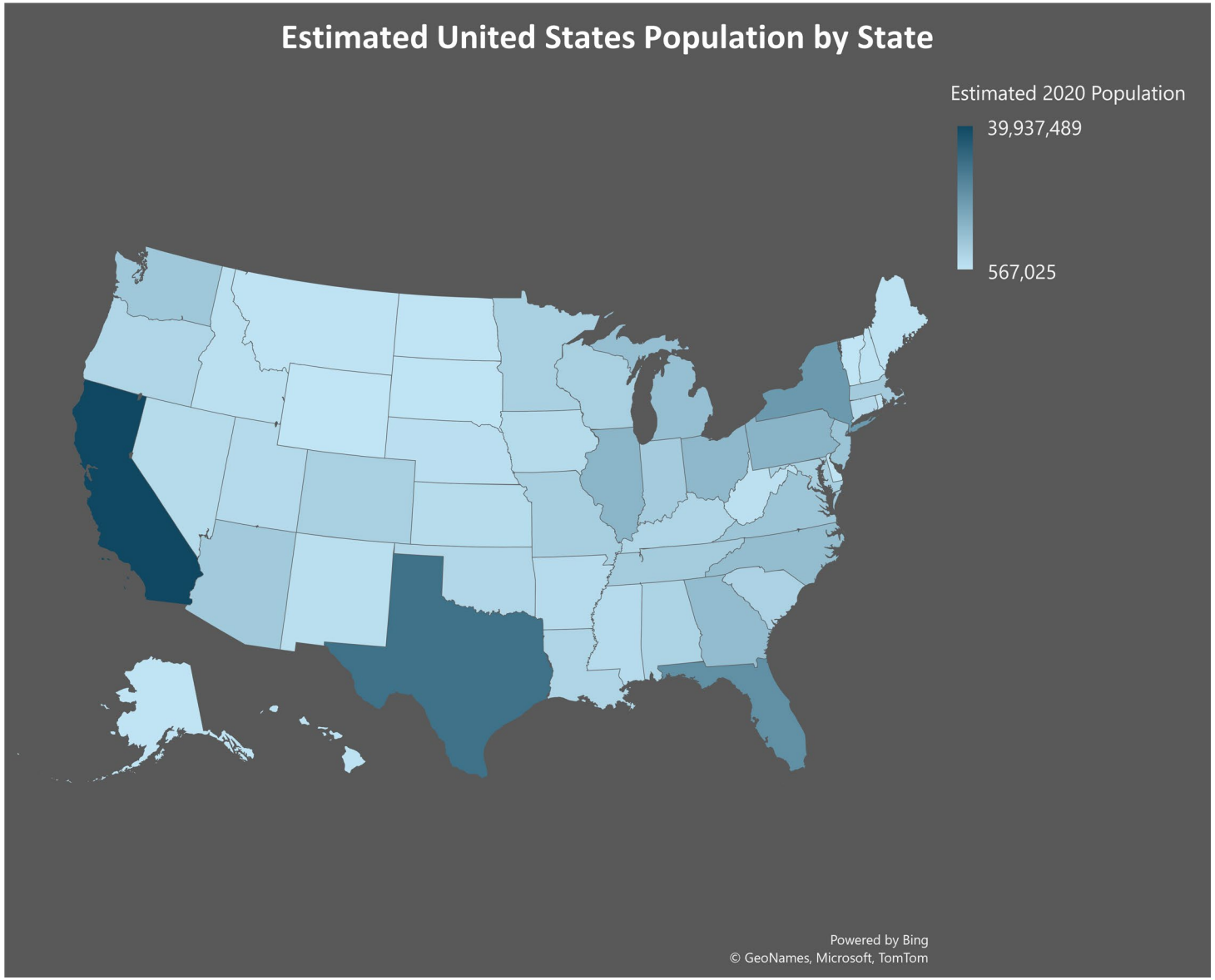


Choropleth Map

Interpretation: This Choropleth map shows the estimated population of each US state in 2020.

Insights

- California was the most populous state in the U.S. in 2020
    - Texas, Florida, and New York were also heavily populated states
- The Northwestern part of the country had less populated states compared to other parts of the country



Heat Map

Interpretation: This heat map compares the change in sales for a store (by %) based on the month and location of the store (by city).

- Blue represents increased sales, where darker shades of blue represent larger sale increases
- Red represents decreased sales, where darker shades of red represent larger decreases
- White represents little to no change in sales

Insights

- Austin was the city with the best overall sales performance over the year
    - April was the only month where Austin did not have the largest increase in sales out of all of the store locations

Store Sales by Month and Location

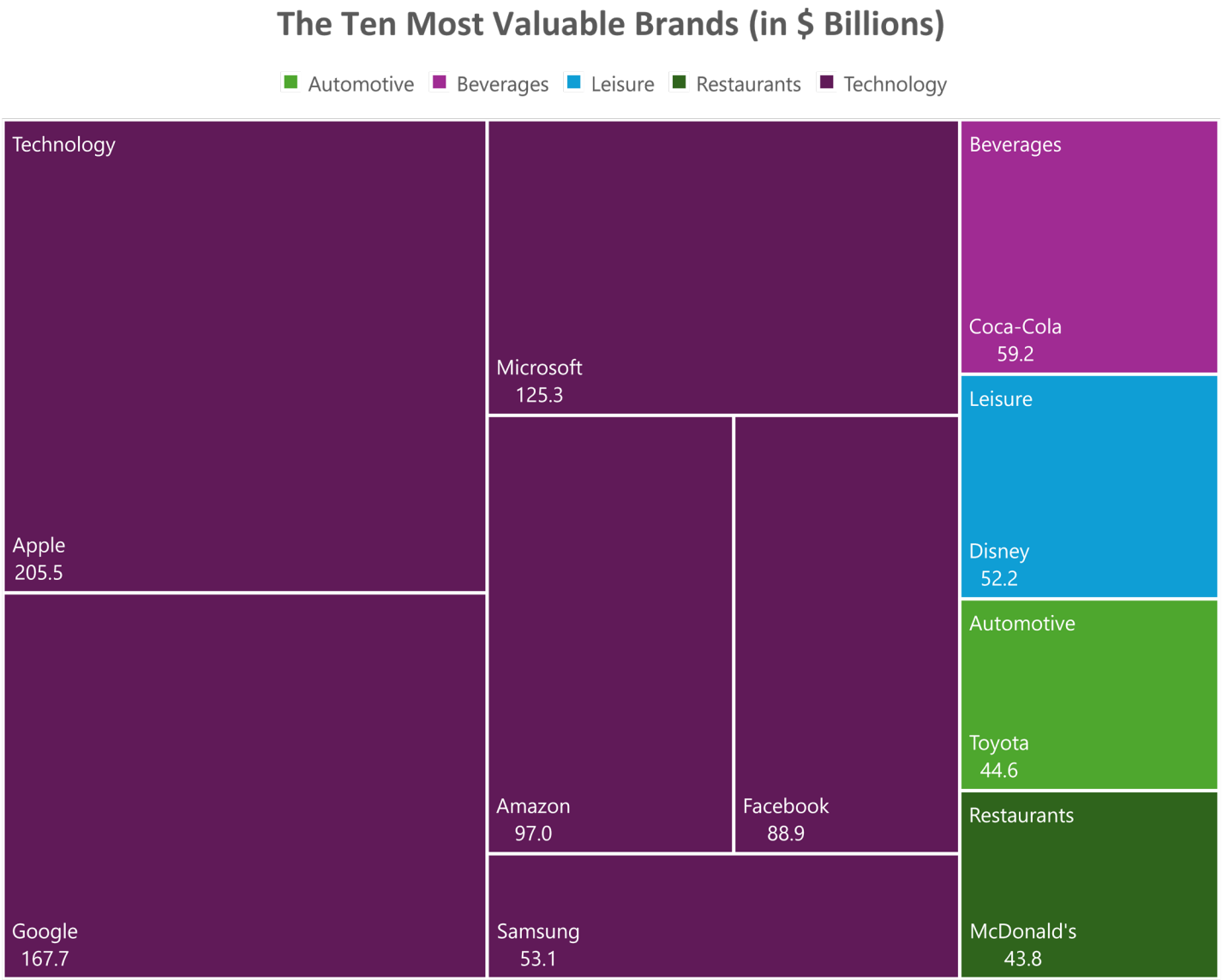
	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
St. Louis	-2%	-1%	-1%	0%	2%	4%	3%	5%	6%	7%	8%	8%
Phoenix	5%	4%	4%	2%	2%	-2%	-5%	-8%	-6%	-5%	-7%	-8%
Albany	-5%	-6%	-4%	-5%	-2%	-5%	-5%	-3%	-1%	-2%	-1%	-2%
Austin	16%	15%	15%	16%	18%	17%	14%	15%	16%	19%	18%	16%
Cincinnati	-9%	-6%	-7%	-3%	3%	6%	8%	11%	10%	11%	13%	11%
San Francisco	2%	4%	5%	8%	4%	2%	4%	3%	1%	-1%	1%	2%
Seattle	7%	7%	8%	7%	5%	4%	2%	0%	-2%	-4%	-6%	-5%
Chicago	5%	3%	2%	6%	8%	7%	8%	5%	8%	10%	9%	8%
Atlanta	12%	14%	13%	17%	12%	11%	8%	7%	7%	8%	5%	3%
Miami	2%	3%	0%	1%	-1%	-4%	-6%	-8%	-11%	-13%	-11%	-10%
Minneapolis	-6%	-6%	-8%	-5%	-6%	-5%	-5%	-7%	-5%	-2%	-1%	-2%
Denver	5%	4%	1%	1%	2%	3%	1%	-1%	0%	1%	2%	3%
Salt Lake City	7%	7%	7%	13%	12%	8%	5%	9%	10%	9%	7%	6%
Raleigh	4%	2%	0%	5%	4%	3%	5%	5%	9%	11%	8%	6%
Boston	-5%	-5%	-3%	4%	-5%	-4%	-3%	-1%	1%	2%	3%	5%
Pittsburgh	-6%	-6%	-4%	-5%	-3%	-3%	-1%	-2%	-2%	-1%	-2%	-1%

Treemap

Interpretation: This treemap shows the ten most valuable brands (in $ billions), color-coded by their industry.

Insights

- Technology is the industry with the most brands among the ten most valuable brands
    - Technology is the only industry with more than one company among the ten most valuable brands

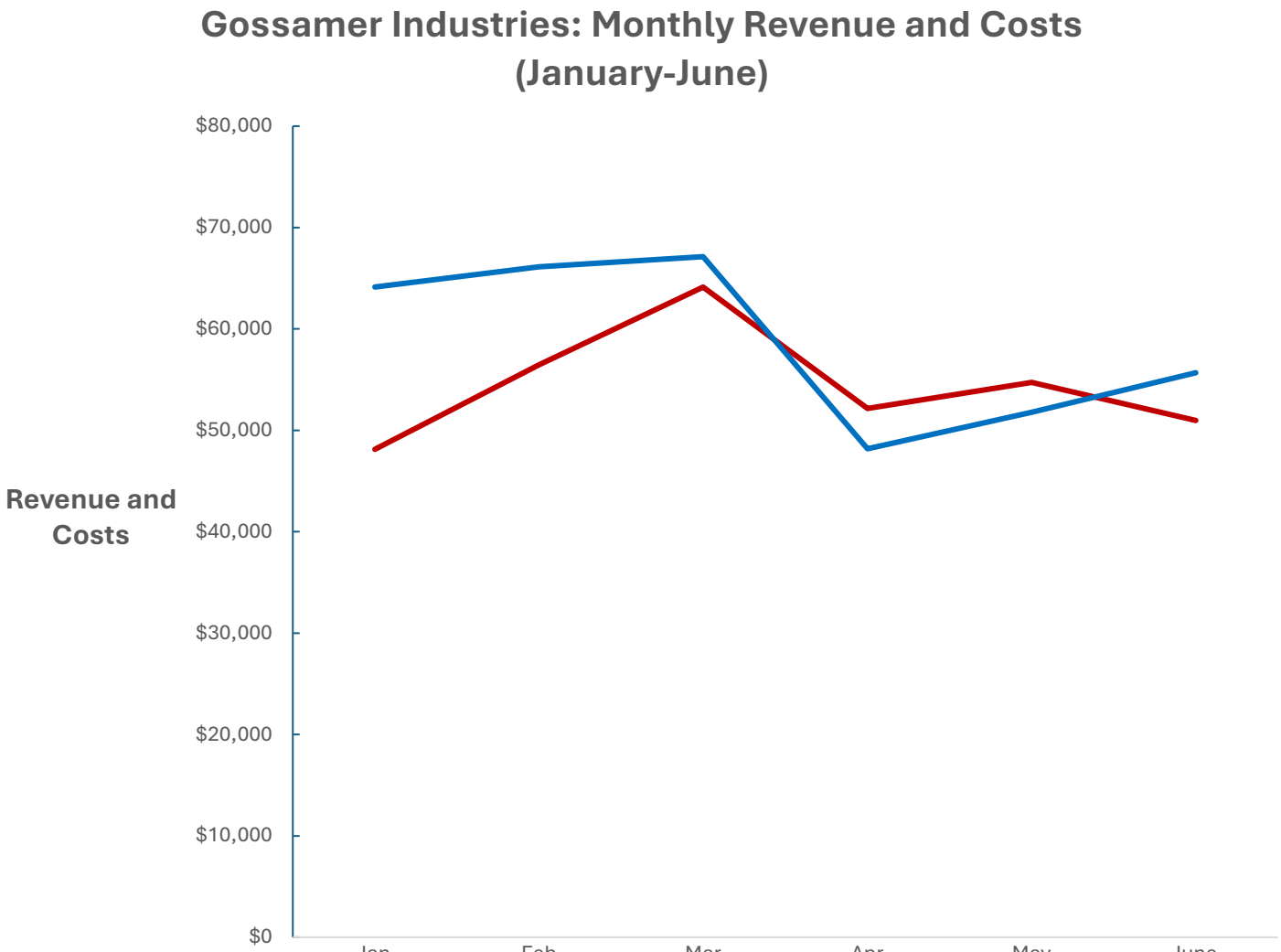


Combined Line Chart and Table

Interpretation: This visualization includes a line chart and a table that shows the revenue and costs (in $) generated by Gossamer Industries every month from January to June.

Insights

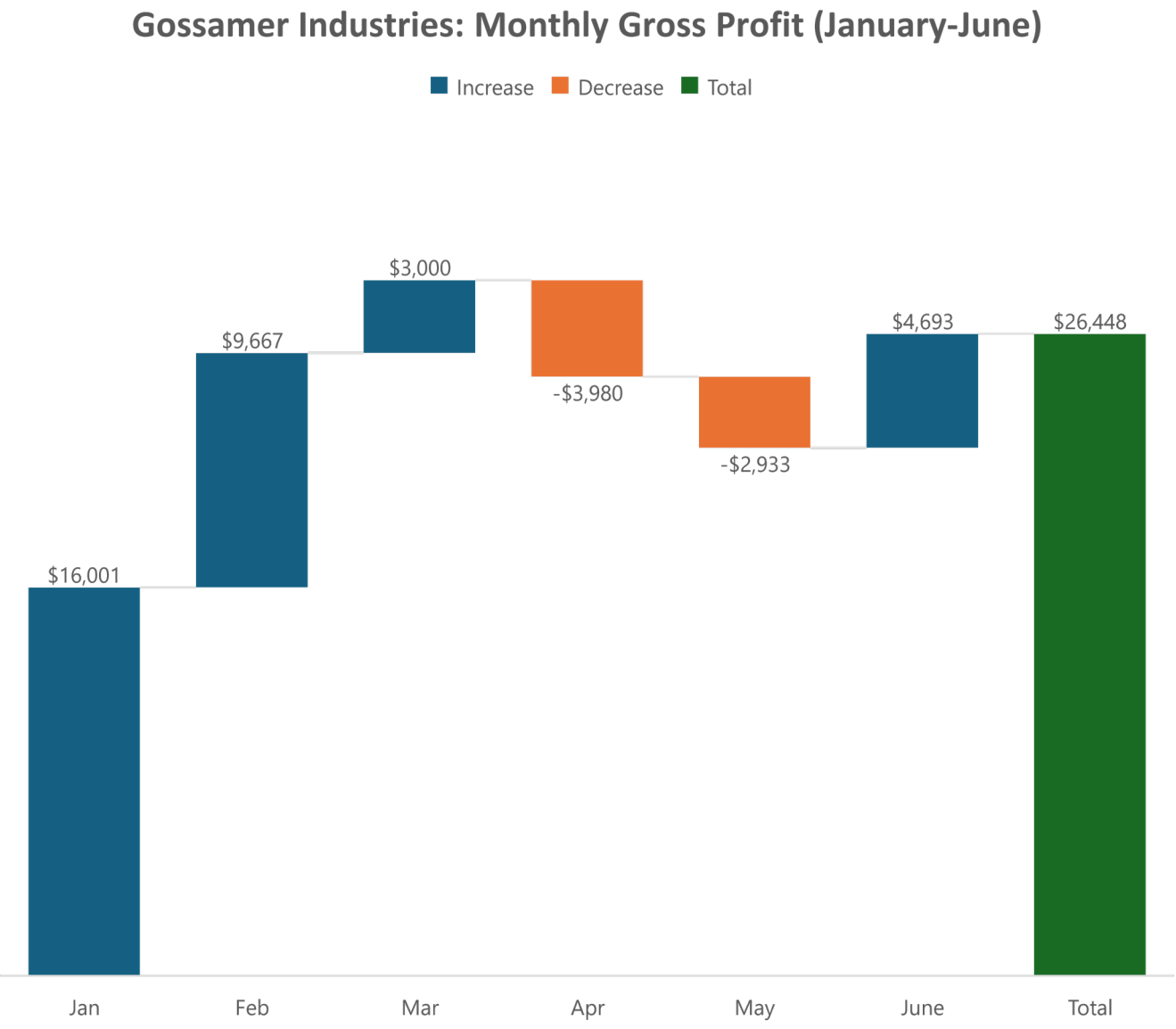
- From January to mid-March, Gossamer Industries made a profit since their revenue exceeded their costs
    - Each month during this timeframe, Gossamer Industries yielded a profit that was smaller than the previous month
- From mid-March to mid-May, Gossamer Industries lost money due to their costs exceeding their revenue
- From mid-May through June, Gossamer Industries started generating a profit again



	Jan	Feb	Mar	Apr	May	June
Costs (in $)	48,123	56,458	64,125	52,158	54,718	50,985
Revenues (in $)	64,124	66,125	67,125	48,178	51,785	55,678

Waterfall Chart

Interpretation: This waterfall chart shows the amount of money Gossamer Industries gained or lost from January to June and the gross profit over the entire six-month period. The color and location of the boxes in the chart represent either a profit or a loss, and the size of the boxes represents the amount of profit or loss.



Insights

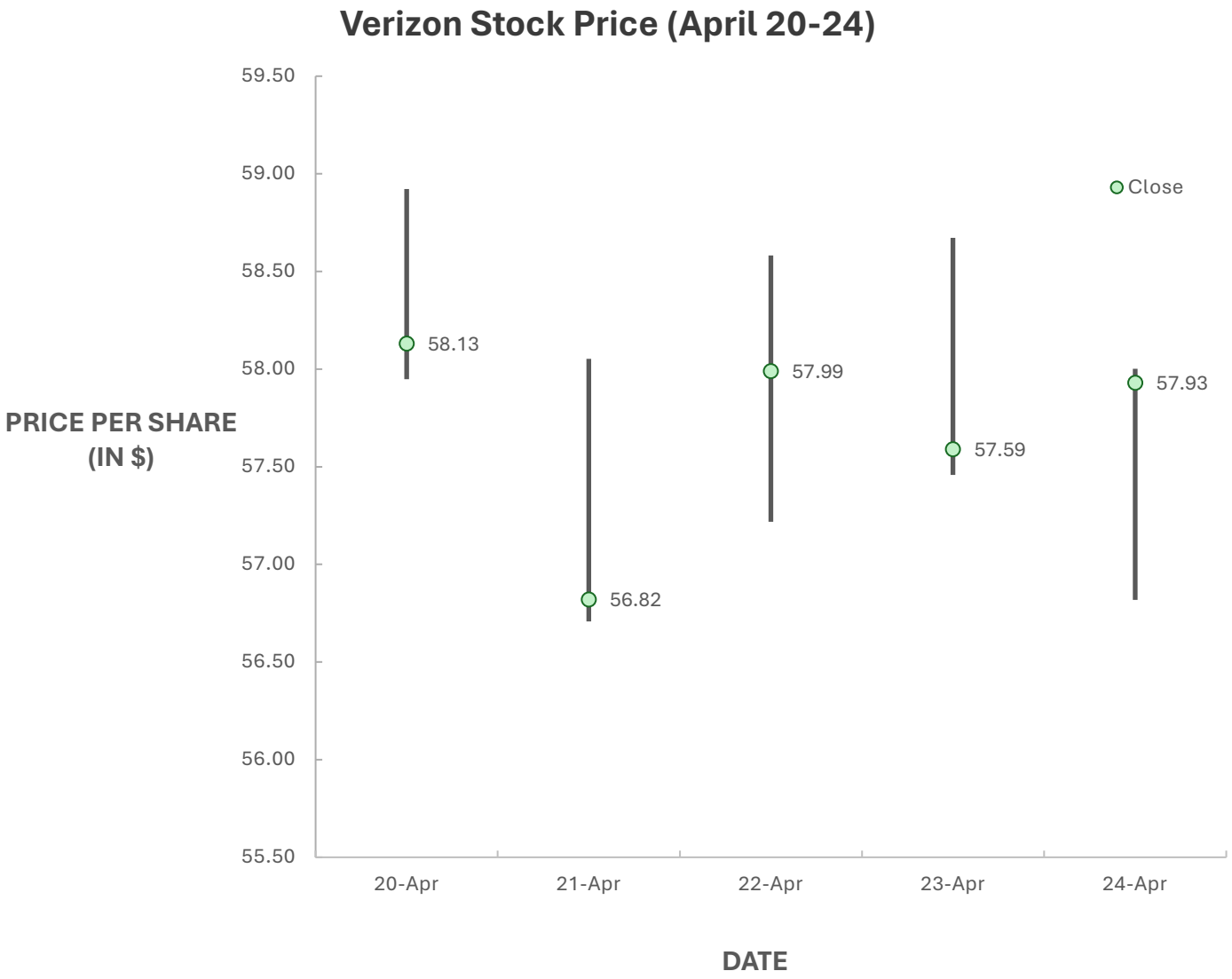
- Gossamer Industries profited from January through March
    - Each month had less profit than the previous month
- Gossamer Industries lost money during April and May
    - The loss in May was less than the loss in April, which is a sign of improvement
- Gossamer Industries profited in June
- Over the six-month period, Gossamer Industries made a profit

Stock Chart

Interpretation: This is a stock chart for Verizon Wireless from April 20th-24th. This chart shows the range of prices per share and the closing price for each day.

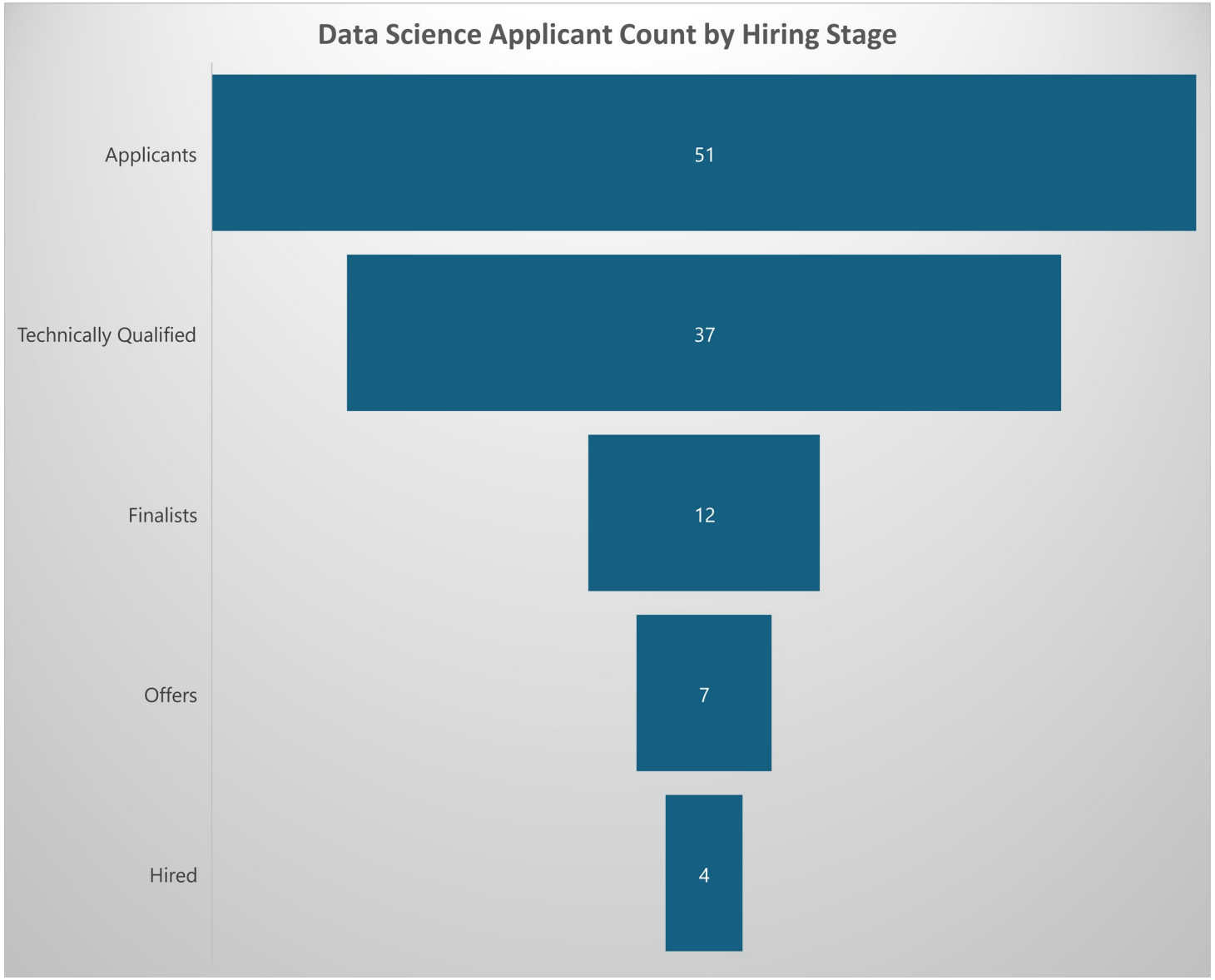
Insights

- April 20th was the day with the highest range of stock prices
- April 21st was the day with the lowest range of stock prices
- Stocks increased on April 22nd -23rd
- With the exception of April 21st, the closing price per share every day was around $58



Funnel Chart

Interpretation: This funnel chart shows the number of data science applicants remaining at each stage of the hiring process.

Insights

- Out of applicants who had the technical skills for a data scientist position, nearly 1/3 of them were selected as finalists
- Out of the finalists, nearly half of them were offered a data science position
- Out of the finalists who were offered a position, nearly half of them were hired
- Out of all of the applicants who applied for a data science position, roughly 13% of them were hired
    - Meaning the position is competitive